# Shape and Space 8 — Vectors · Column vectors · Vector geometry

Calculator allowed. Marks in brackets. ★ = harder question. Bold letters are vectors; $|\mathbf{a}|$ is the magnitude (length) of $\mathbf{a}$.

**1** $\mathbf{p} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$, $\mathbf{q} = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$, $\mathbf{r} = \begin{pmatrix} 1 \\ 5 \end{pmatrix}$ and $2\mathbf{p} + \mathbf{r} = \mathbf{q} - \mathbf{s}$. [5]

**a** Work out $\mathbf{s}$ as a column vector. **b** Calculate $|\mathbf{s}|$ to 1 d.p.

**a** $\mathbf{s} = \mathbf{q} - 2\mathbf{p} - \mathbf{r} = \begin{pmatrix} -2-6-1 \\ 4+2-5 \end{pmatrix} = \begin{pmatrix} -9 \\ 1 \end{pmatrix}$. **b** $|\mathbf{s}| = \sqrt{(-9)^2 + 1^2} = \sqrt{82} = 9.1$ .

**2** Points A and B have position vectors $\overrightarrow{OA} = \begin{pmatrix} 4 \\ 7 \end{pmatrix}$ and $\overrightarrow{OB} = \begin{pmatrix} 16 \\ 12 \end{pmatrix}$. [8]

**a** Write down $\overrightarrow{AB}$. **b** Calculate $|\overrightarrow{AB}|$.

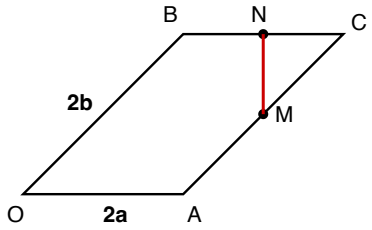
**c** Find the angle $\overrightarrow{AB}$ makes with the $x$-axis, to 1 d.p.

**d** X is the mid-point of AB. Write $\overrightarrow{OX}$ as a column vector.

**a** $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 12 \\ 5 \end{pmatrix}$. **b** $\sqrt{12^2 + 5^2} = \sqrt{169} = 13$. **c** $\tan\theta = \dfrac{5}{12} \Rightarrow \theta = 22.6°$. **d** $\overrightarrow{OX} = \overrightarrow{OA} + \frac{1}{2}\overrightarrow{AB} = \begin{pmatrix} 4+6 \\ 7+2.5 \end{pmatrix} = \begin{pmatrix} 10 \\ 9.5 \end{pmatrix}$ (the mean of the two position vectors).

**3** [9]

OACB is a parallelogram with $\overrightarrow{OA} = \mathbf{2a}$ and $\overrightarrow{OB} = \mathbf{2b}$. M is the mid-point of AC and N is the mid-point of BC.




Express in terms of $\mathbf{a}$ and $\mathbf{b}$: **a** $\overrightarrow{OC}$ **b** $\overrightarrow{OM}$ **c** $\overrightarrow{ON}$ **d** $\overrightarrow{NM}$

**e** What do your answers tell you about NM and BA?

**a** $\overrightarrow{OC} = \overrightarrow{OA} + \overrightarrow{AC} = \mathbf{2a} + \mathbf{2b}$ (opposite sides of a parallelogram are equal vectors, $\overrightarrow{AC} = \overrightarrow{OB}$).

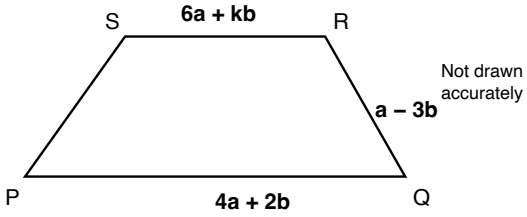
**b** $\overrightarrow{OM} = \overrightarrow{OA} + \frac{1}{2}\overrightarrow{AC} = \mathbf{2a} + \mathbf{b}$. **c** $\overrightarrow{ON} = \overrightarrow{OB} + \frac{1}{2}\overrightarrow{BC} = \mathbf{2b} + \mathbf{a}$.

**d** $\overrightarrow{NM} = \overrightarrow{OM} - \overrightarrow{ON} = (\mathbf{2a} + \mathbf{b}) - (\mathbf{a} + \mathbf{2b}) = \mathbf{a} - \mathbf{b}$.

**e** $\overrightarrow{BA} = \overrightarrow{OA} - \overrightarrow{OB} = \mathbf{2a} - \mathbf{2b} = 2(\mathbf{a} - \mathbf{b}) = 2\,\overrightarrow{NM}$. So NM is parallel to BA and half its length.

**4** [3]

PQRS is a trapezium with PQ parallel to SR. $\overrightarrow{PQ} = \mathbf{4a} + \mathbf{2b}$, $\overrightarrow{QR} = \mathbf{a} - \mathbf{3b}$ and $\overrightarrow{SR} = \mathbf{6a} + k\mathbf{b}$, where $k$ is a constant. Find $k$, showing your working.




Parallel vectors are multiples of each other: $\overrightarrow{SR} = \lambda\,\overrightarrow{PQ}$, so $\mathbf{6a} + k\mathbf{b} = \lambda(\mathbf{4a} + \mathbf{2b})$. Comparing $\mathbf{a}$ parts: $4\lambda = 6 \Rightarrow \lambda = 1.5$. Comparing $\mathbf{b}$ parts: $k = 2\lambda = 3$.

**Total: 25 marks**

Check: add vectors component by component · $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$ (end minus start) · magnitude by Pythagoras · a route around a shape: go along known vectors, backwards means a minus sign · parallel ⇔ one vector is a multiple of the other; same multiple through a shared point ⇔ collinear.